ARM Accredited Engineer

Mock Test 3 - Answers

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Instructions

This mock test is designed to give prospective test-takers an opportunity to sample questions of a similar scope and level of difficulty to those included in the live AAE certification test.

The test consists of 10 multiple choice questions and an accompanying document provides answers to these questions along with a rationale for each question and answer.

We suggest that you allow yourself 10 minutes to complete this test, without the use of any reference materials or learning materials.

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Question 1

Which one of the following ARM processors contains a Snoop Control Unit (SCU), for hardware cache coherency?

A) Cortex-A8
B) Cortex-M3
C) Cortex-R4
D) Cortex-A5 MPCore ✓

Answer D is correct. The Snoop Control Unit (SCU) maintains coherency between the individual L1 data caches in the ARM MP processors.

Question 2

Which one of the following statements is TRUE for hardware breakpoints?

A) Hardware breakpoints utilize the BPKT instruction on ARM processors
B) Hardware breakpoints are not suitable for debugging exception handlers
C) Hardware breakpoints can be used to debug code running from read-only memory ✓
D) Cache maintenance operations may be required when placing a hardware breakpoint

Answer C is correct. ARM DS-5 Using the Debug hardware Configuration Utilities: Hardware instruction breakpoints do not require the instruction in memory to be changed. This means that they can be used to debug code in Flash and ROM, and can be used with self-modifying code.

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Question 3

Address	Contents
0x24	0x06
0x25	0xFC
0x26	0x03
0x27	0xFF

If r0 has the value 0x24, what is the content of r12 after executing the following instruction?

LDRB r12, [r0], #2

A) 0xFC
B) 0x03
C) **0x06** ✓
D) 0xFF

Answer C is correct. The instruction LDRB has a post-indexed address mode for the base register, which means that a byte (0x06) will be loaded from the address in r0 into r12, and then the address in r0 will be updated with the offset (#2).

Question 4

What is the significance of "!" in a load/store instruction?

E) Don't update base register in post-indexed load/store
F) Don't update base register in pre-indexed load/store
G) Update base register in post-indexed load/store
H) **Update base register in pre-indexed load/store** ✓

Answer D is correct. ARM Architecture Reference Manual section A5.2.5 Load and Store Word or Unsigned Byte:

! – Sets the W bit, causing base register update.

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Question 5

For ARMv7-A memory management, which attribute control field is used in a page table entry to control use of a page table with a given Address Space IDentifier (ASID)?

A) AP (Access Permission)
B) **nG (Not Global)** ✓
C) SH (Shared)
D) XN (Execute Never)

Answer B is correct. ARM Architecture Reference Manual for v7-AR section B3.9.1 Global and process-specific translation table entries:

nG == 0 The translation is global, meaning the region is available for all processes.
nG == 1 The translation is non-global, or process-specific, meaning it relates to the current ASID, as defined by the CONTEXTIDR.

Question 6

Which TWO of these statements are true for a function that has been built to use hardfp? (Please select TWO options)

A) The function must not read from or write to the stack
B) **The result of the function can be returned in a VFP register** ✓
C) Floating point function arguments can be passed in core registers
D) Floating point calculations are performed using the NEON unit only
E) **Up to 16 function arguments can be passed in floating point registers** ✓

Answers B and E are correct. AAPCS (Procedure Call Standard for ARM Architecture) section 5.1.2.1 VFP register usage conventions: registers s0-s15 (d0-d7, q0-q3) do not need to be preserved (and can be used for passing arguments or returning results in standard procedure-call variants).

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Question 7

When a linker creates a static image:

A) it records the entry point in the ELF header. ✓
B) it places the entry point at the lowest address.
C) the entry point must be given on the command-line.
D) the entry point cannot be given on the command-line.

Answer A is correct. ARM Linker Reference Manual: The image can contain multiple entry points, but the initial entry point specified with this option is stored in the executable file header for use by the loader.

Question 8

Which of the following provides fastest access for the processor?

A) Tightly Coupled Memory (TCM)
B) Hard disk
C) Onboard flash memory
D) Register File ✓

Answer D is correct. Register access is always faster than any memory access.

Question 9

What are software-generated interrupts in a Generic Interrupt Controller (GIC) generally used for?

A) Causing a delay
B) Entering a low power state
C) Communicating between processors ✓
D) Calling an operating system function

Answer C is correct. Cortex-A Series Programmer's Guide section 23.3 Handling interrupts in an SMP system: SGI (Software-generated Interrupt) register can assert private software generated interrupt on any core, or a group of cores. These Inter-processor Interrupts can be used for kernel synchronization operations, or for communicating between AMP processors.

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Question 10

Which of the following methods could be used to calculate the Cycles Per Instruction (CPI) value for a portion of code?

A) Single-step the code using a JTAG debugger
B) Count the cache hits
C) **Use the PMU event counters** ✓
D) Time it with a stopwatch

Answer C is correct. Cortex-A Series Programmer's Guide section 16.1.4 ARM performance monitor: The performance monitor hardware is able to count several events, using multiple counters. Normally, we combine together multiple values to generate useful parameters to optimize. For example, we can choose to count the total number of clock cycles and the number of instructions executed and use this to derive a cycles per instruction figure which is a useful proxy for the efficiency with which the processor is operating.

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